HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F78AA1

ACTUARIAL AND FINANCIAL MATHEMATICS A

18 December 2008

Time allowed 2 hours

Attempt all 7 questions

Total Marks 60

Only approved calculators may be used

NUMERICAL SOLUTIONS

1. (a) Bookwork.

   (b) (i) 63.0025
   (ii) 10.0000
   (iii) 10.0090
   (iv) 2.36900.

2. (a) 96.01606 = £96.02 rounded.

   (b) 145.46456 = £145.46 rounded.

3. (a) (i) £277.55.
   (ii) £11,818.17.

   (b) Solving $11818.17 \leq 277.55\ a_{\overline{t}|}^{0.007}$ to find $t = 50.735$, the loan must be extended by 3 months.

4. £65.86.

5. (a) Check that:

$$200,000(1.12)^{-6} + 500,000(1.12)^{-2}a_{\overline{4}|}^{0.12} - 1,000,000 = 312,004.$$

   (b) The equation of value, expressed at $t = 6$, is:

$$f(i) = -1000(1+i)^6 + 500s_{\overline{4}|}^{i} + 200 = 0.$$

   Find $f(0.19) = +5.87$ and $f(0.20) = -101.98$ and interpolate linearly to get $i^* \approx 0.1905$.

   (c) The loan is repaid at $t = 5$, and the accumulated profit is £963,717.

6. (a) £1.7608.

   (b) £150.6503.

   (c) Choose time unit 1/2 year. The equation of value as at redemption ($t = 20$) is:

$$\begin{aligned} f(i) &= 110(1+i)^{20} \\ &\quad - \frac{3}{2} \times 1.484239 \times \left[\frac{(1+i)^{19}}{1.012423^{19}} + \frac{(1+i)^{18}}{1.012423^{18}} + \cdots + 1 \right] \\ &\quad - 148.4239 = 0 \end{aligned}$$

   The term is square brackets is $s_{\overline{20}|}^{j}$ where $1 + j = (1+i)/1.012423$. At 6% p.a., $f(0.029563) = -3.90$ while at 7% p.a., $f(0.034408) = +12.93$. Hence the monetary yield lies between 6% p.a. and 7% p.a..

   (d) Cash income increases, so monetary yield must increase too.

7. In 2007–8 the equation of value for the yield i_1 is:

$$f(i_1) = 80(1+i_1)^2 + 20(1+i_1) - 110 = 0.$$

Solve this quadratic to get $i_1 = 0.054248$. In 2009 the equation of value for the yield is:

$$f(i_2) = 90(1+i_2) - 100 = 0$$

which has solution $i_2 = 0.111111$. Hence the LIRR satisfies:

$$(1+\text{LIRR})^3 = (1+i_1)^2(1+i_2) \Rightarrow \text{LIRR} = 0.072871.$$